AP CALCULUS AB

Unit 1 • Lesson 1.11 practice • Answer key

Lesson 1.11 • Defining Continuity at a Point

4 multiple-choice questions and 1 free-response question • Answer key with explanations

Question 1

Let $f(x) = x^2$ for $x \neq 2$ and $f(2) = 1$. Which condition for continuity at $x = 2$ fails?

(A) $f(2)$ is defined.
(B) $\lim_{x \to 2} f(x)$ exists.
(C) $\lim_{x \to 2} f(x) = f(2)$.
(D) None; f is continuous at 2.

Solution

Answer: (C). The limit is 4 but $f(2) = 1$.

Question 2

Let $f(x) = \begin{cases} 2x + 1, & x < 1 \\ 3, & x = 1 \\ x^2 + 2, & x > 1. \end{cases}$ Is f continuous at $x = 1$?

(A) Yes
(B) No, $f(1)$ is undefined.
(C) No, the limit does not exist.
(D) No, the limit is not equal to $f(1)$.

Solution

Answer: (A). Left limit 3, right limit 3, and $f(1) = 3$.

Question 3

Why is $f(x) = \dfrac{x^2 - 9}{x - 3}$ not continuous at $x = 3$?

(A) The limit does not exist.
(B) $f(3)$ is undefined.
(C) The limit is infinite.
(D) It is continuous at $x = 3$.

Solution

Answer: (B). The denominator is 0 at $x = 3$, so $f(3)$ is undefined (the limit is 6).

Question 4

Which function is continuous at $x = 0$?

(A) $\dfrac{1}{x}$
(B) $\dfrac{|x|}{x}$
(C) $\dfrac{\sin x}{x}$
(D) $|x|$

Solution

Answer: (D). $|x|$ is defined at 0 and its limit there is $0 = |0|$. The others are undefined at 0.

Question 5 (Free response)

Let $f(x) = \begin{cases} x^2 - 1, & x < 2 \\ 3, & x = 2 \\ 5 - x, & x > 2. \end{cases}$

(a) Find $\lim_{x \to 2} f(x)$, showing both one-sided limits.
(b) Is f continuous at $x = 2$? Justify using the definition of continuity.
(c) Is f continuous at $x = 0$? Explain.

Solution
(a) $\lim_{x \to 2^-} f(x) = 4 - 1 = 3$ and $\lim_{x \to 2^+} f(x) = 5 - 2 = 3$, so $\lim_{x \to 2} f(x) = 3$.
(b) Yes: $f(2) = 3$ is defined, $\lim_{x \to 2} f(x) = 3$ exists, and $\lim_{x \to 2} f(x) = f(2)$.
(c) Yes. Near $x = 0$, $f(x) = x^2 - 1$, a polynomial, which is continuous everywhere.